Helical Time, Spiral Arithmetic, and the Geometry of Couplings

A self-contained and intuitive presentation

September 17, 2025

Abstract

We develop a self-contained, geometry-first framework in which (i) a single normalized logarithmic spiral carries a complete ordered-field structure $(S, \oplus, \otimes, \leq_S)$ isomorphic to $(\mathbb{R}, +, \cdot, \leq)$; (ii) a helical internal geometry (a compact angle θ fibered over log-radius $\mu = \ln r$) encodes gauge couplings as fiber radii and fermion Yukawa matrices as overlap integrals of localized helical zero-modes; and (iii) a simple "tick-stability" principle—actualization occurs near integer ticks in the spiral–log coordinate—explains hierarchical masses, small mixings, and predicts log-periodic spectral signatures. We keep the presentation intuitive, with minimal formalism sufficient for clarity, and collect all definitions, normalizations, and numerical checks in appendices.

Contents

1 Executive summary (one page) **1**

2 Spiral arithmetic: numbers as points on a logarithmic spiral **2**
- 2.1 Normalized spiral and similarity group . . . 2
- 2.2 Spiral exponential and logarithm . . . 2
- 2.3 Arithmetic and order on the spiral . . . 2

3 Tick-stability principle and actualization **2**
- 3.1 Principle . . . 2
- 3.2 Why this is natural . . . 2

4 Helical kinematics in one page (for intuition) **3**

5 Gauge sector from helical fibers **3**
- 5.1 Electromagnetism ($U(1)$) and α . . . 3
- 5.2 Electroweak mixing angle from a radius ratio . . . 3
- 5.3 Color ($SU(3)$) and α_s . . . 4

6 Fermion masses and mixings from helical overlaps **4**
- 6.1 Profiles, levels, windings . . . 4
- 6.2 Two concrete benchmarks . . . 4

7 Predictions and constraints **5**
- 7.1 Derived scales and cross-checks . . . 5

8 Cosmological note (one paragraph and a remedy) **5**

9 Experimental roadmap **5**

10 Conclusions **6**

A Units, symbols, and constants **6**

B Spiral arithmetic: proofs and variations **6**

C Golden hierarchies and level counts **6**

D Minimal numerical integrator (illustrative) **6**

E Yukawa overlaps: symmetric-width corollary **7**

F IGS clock tests: φ-comb detection in satellites and stations **7**
F.1 Results with diurnals removed (recommended analysis) 7
F.2 Results with diurnals kept (control) 8
F.3 What this constrains 8
F.4 Figures (quick-look diagnostics) 8

1 Executive summary (one page)

Idea. The physics of scales and phases can be encoded on a single normalized logarithmic spiral with a one-parameter group of similarities. The spiral admits canonical "ticks" at integer steps of a chosen unit angle and radius-ratio. We transfer ordinary arithmetic from $\mathbb{R}$ to points on the spiral, and postulate a *tick-stability principle*: potential modes become actual (long-lived, resonant) when their spiral–log coordinates lie within a small window around integer ticks and helical windings satisfy a simple selection rule.

Outcomes.

- *Gauge sector from geometry.* A compact helical fiber for $U(1)$, $SU(2)$, $SU(3)$ with radii R_Y, R_2, R_3 produces couplings via $1/g_a^2 = \kappa_a M_*^2 R_a^2$ (with order-one κ_a). The fine-structure constant α fixes the $U(1)$ fiber radius $R_\theta \sim \mathcal{O}(10)\,\ell_P$; electroweak mixing is the ratio R_Y/R_2; the strong coupling fixes $R_3 \sim \mathcal{O}(10)\,\ell_P$ at high scale.
- *Fermion masses and mixing.* Chiral zero-modes are localized packets in μ with integer/half-integer tick centers and simple integer windings along θ. Triple overlaps (left, right, Higgs) generate hierarchical Yukawas; small mixings arise from slight left-handed misalignment and near-neighbor winding differences. We give tuned benchmark sets.
- *Predictions.* (i) Log-periodic spectral combs: preferred frequencies $f_n = f_0\,\lambda^n$ (or times $\tau_n = \tau_0\,\lambda^n$) with λ the spiral ratio (golden case $\lambda = \varphi$). (ii) Standard probes constrain the helical scale: photon time-of-flight dispersion, GW speed/dispersion, and precision clocks.

Scope and humility. The spiral arithmetic itself is a coordinate choice; the new physics enters through the tick-stability postulate and the helical compact fiber. We keep explicit "order-one" factors κ_a and list numerical consequences for transparent comparison with data.

2 Spiral arithmetic: numbers as points on a logarithmic spiral

2.1 Normalized spiral and similarity group

Fix parameters: a radius ratio $\lambda > 1$ per unit angle, a unit angle $\alpha_0 > 0$, and a base radius $r_0 > 0$. Let R_θ be rotation by θ and D_s dilation by $s > 0$. Define the one-parameter similarity group

$$G := \{g_t : \mathbb{R}^2 \to \mathbb{R}^2 \mid g_t = D_{\lambda^t} \circ R_{t\alpha_0}\}_{t\in\mathbb{R}}, \qquad g_s \circ g_t = g_{s+t},\ g_0 = \mathrm{id}. \tag{1}$$

Let $e = (r_0, 0)$ and $S := \{g_t(e) : t \in \mathbb{R}\}$. In polar coordinates $S = \{(r(\theta), \theta)\}$ with $r(\theta) = r_0\, \lambda^{\theta/\alpha_0}$.

2.2 Spiral exponential and logarithm

The action of G on S is free and transitive. Define the bijections

$$\mathrm{Exp} : \mathbb{R} \to S, \quad \mathrm{Exp}(t) = g_t(e); \qquad \mathrm{Log} : S \to \mathbb{R}, \quad \mathrm{Log}(x) = t \iff x = g_t(e). \tag{2}$$

Covariance: $\mathrm{Log}(g_t(x)) = \mathrm{Log}(x) + t$, $\mathrm{Exp}(a+b) = g_a(\mathrm{Exp}(b))$. We call t the *spiral–log coordinate* of x.

2.3 Arithmetic and order on the spiral

Define for $x, y \in S$:

$$x \oplus y := \mathrm{Exp}(\mathrm{Log}x + \mathrm{Log}y), \qquad x \otimes y := \mathrm{Exp}((\mathrm{Log}x)(\mathrm{Log}y)), \tag{3}$$

with identities $0_S = \mathrm{Exp}(0) = e$ and $1_S = \mathrm{Exp}(1) = g_1(e)$. Inverses are $\ominus x = \mathrm{Exp}(-\mathrm{Log}x)$ and $x^{\otimes -1} = \mathrm{Exp}((\mathrm{Log}x)^{-1})$ for $x \neq 0_S$. Order: $x \leq_S y \iff \mathrm{Log}x \leq \mathrm{Log}y$. Equivalently,

$$x \oplus y = g_{\mathrm{Log}x}(y) = g_{\mathrm{Log}y}(x), \qquad x \otimes y = g_{(\mathrm{Log}x)(\mathrm{Log}y)}(e). \tag{4}$$

Theorem 1 (Field structure)**.** *$(S, \oplus, \otimes, \leq_S)$ is an ordered field, and $\mathrm{Exp} : (\mathbb{R}, +, \cdot, \leq) \to (S, \oplus, \otimes, \leq_S)$ is an order–field isomorphism.*

Remark 1 (Ticks and golden specialization)**.** Ticks *are the points $\mathrm{Exp}(n) = g_n(e)$, $n \in \mathbb{Z}$. Setting $\lambda = \varphi = (1+\sqrt{5})/2$ and e.g. $\alpha_0 = \frac{\pi}{2}$ produces radii $r_0\varphi^n$ at quarter-turn ticks. Zeckendorf/Fibonacci numeration furnishes a canonical digit system for the monoid $(\mathbb{N}_S, \oplus)$.*

3 Tick-stability principle and actualization

3.1 Principle

Tick-stability (postulate). There exists a tick window of half-width $\delta \in (0, \frac{1}{2}]$ in the spiral coordinate $\mathcal{T}_\delta := \bigcup_{n\in\mathbb{Z}} [n-\delta, n+\delta]$ such that modes with $t \in \mathcal{T}_\delta$ are dynamically preferred (long-lived or resonant). Actualization of a process involving three wave packets (initial, final, mediator) requires their coordinates to fall in overlapping tick windows and satisfy a helical winding selection rule (below).

3.2 Why this is natural

For Gaussian profiles localized at μ-centers μ_i^Q, μ_j^R, μ_H with widths $(\sigma_Q, \sigma_R, \sigma_H)$ (here $\mu = \ln r$ is proportional to t), the triple overlap scales as

$$|Y_{ij}^R| \;\propto\; \exp\left[-\tfrac{1}{2}\left(\frac{(\mu_i^Q - \bar\mu)^2}{\sigma_Q^2} + \frac{(\mu_j^R - \bar\mu)^2}{\sigma_R^2} + \frac{(\mu_H - \bar\mu)^2}{\sigma_H^2}\right)\right], \quad \bar\mu = \frac{\mu_i^Q/\sigma_Q^2 + \mu_j^R/\sigma_R^2 + \mu_H/\sigma_H^2}{\sigma_Q^{-2} + \sigma_R^{-2} + \sigma_H^{-2}}. \tag{5}$$

This is maximized when all three centers coincide; restricting centers to ticks (or small neighborhoods) therefore maximizes couplings. In discrete-scale dynamics (Floquet intuition) stability bands likewise occur at $\log f \in \log f_0 + \mathbb{Z}\log\lambda$.

Remark 2 (Practical mapping of δ to widths)**.** *With $\mu = \mu_0 + n\Delta$, $\Delta = \ln\lambda$, a common width σ implies a tolerance $\delta \sim \sigma/\Delta$. Non-detection of log-periodic spectral combs constrains δ and hence the geometric widths.*

4 Helical kinematics in one page (for intuition)

We work in natural units $c = \hbar = 1$ unless noted. The helical manifold has a compact angle $\theta \sim \theta + 2\pi$ fibered over log-radius $\mu = \ln r$. A minimal kinematic ansatz for a worldline parameter s is

$$\frac{dr}{ds} = \beta\, r\left(1 - \frac{r}{r_{\max}}\right), \qquad \frac{d\theta}{ds} = \omega_0\, r^{-\alpha}, \qquad \frac{d\vec{x}}{ds} = v_z(r)\,\hat{\mathbf{z}}, \quad v_z(r) = v_\infty\Big(1 - e^{-r/r_*}\Big), \tag{6}$$

with α dimensionless, $[\beta] = T^{-1}$, $r_{\max} > 0$, $0 < v_\infty \le 1$. This produces a regular spiral at small r, saturating longitudinal speed at large r. We identify $\mu = \ln r$ (hence $t \propto \mu$) and use ticks in μ to label preferred levels.

5 Gauge sector from helical fibers

We take gauge bosons as Kaluza–Klein connections along compact helical fibers with radii R_a and order-one geometric normalizations κ_a. Dimensional reduction gives

$$\frac{1}{g_a^2(\mu_H)} = \kappa_a\, M_*^2\, R_a^2, \qquad \alpha_a(\mu_H) = \frac{g_a^2(\mu_H)}{4\pi} = \frac{1}{4\pi\,\kappa_a\, M_*^2 R_a^2}, \tag{7}$$

at a high matching scale $\mu_H \sim 1/R_a$ (we use the reduced Planck mass $\bar{M}_{\rm Pl} \approx 2.435 \times 10^{18}$ GeV as reference).

5.1 Electromagnetism ($U(1)$) and α

With $\kappa_1 = \frac{1}{2}$ one finds

$$\alpha \;=\; \frac{1}{2\pi\, M_*^2 R_\theta^2} \quad \Rightarrow \quad R_\theta = \frac{1}{\sqrt{2\pi\alpha}\, M_*}\,. \tag{8}$$

Numerically, with $\alpha^{-1} \simeq 137.036$ and $M_* = \bar{M}_{\rm Pl}$, $R_\theta \simeq 3.8 \times 10^{-34}\,\mathrm{m} \approx 24\,\ell_P$. Different κ_1 or using the unreduced $M_{\rm Pl}$ shift this by order-unity, but the radius remains $\mathcal{O}(10)\,\ell_P$.

5.2 Electroweak mixing angle from a radius ratio

Let R_Y (hypercharge) and R_2 ($SU(2)_L$) be the two fiber radii with normalizations κ_Y, κ_2. Then

$$\sin^2\theta_W = \frac{g'^2}{g'^2 + g^2} = \frac{1}{1 + \lambda\, R_Y^2/R_2^2}, \qquad \lambda := \frac{\kappa_Y}{\kappa_2} \times (\text{hypercharge norm}). \tag{9}$$

Two simple choices reproduce $\sin^2\theta_W(m_Z) \approx 0.231$:

- **GUT-normalized hypercharge:** take $\lambda = 5/3$, $R_Y^2/R_2^2 = 2 \Rightarrow \sin^2\theta_W = 3/13 = 0.23077$.
- **Golden-step ratio:** $\lambda = 1$, $R_Y/R_2 = \varphi^{5/4} \Rightarrow \sin^2\theta_W = 1/(1 + \varphi^{5/2}) \approx 0.23094$.

These realize the weak mixing angle as a *pure geometric ratio* of helical radii (ticks or half-ticks).

5.3 Color ($SU(3)$) and α_s

Equation (7) gives

$$R_3 = \frac{1}{M_*\,\sqrt{\kappa_3}\, g_s(\mu_H)} = \frac{1}{M_*\,\sqrt{4\pi\kappa_3\,\alpha_s(\mu_H)}}. \tag{10}$$

Running $\alpha_s(m_Z) \approx 0.118$ to high scale ($\mu_H \sim M_*$) at one loop yields $\alpha_s(\mu_H) \sim 0.02$ and $g_s(\mu_H) \sim 0.50$, giving $R_3 \sim (7{-}14)\,\ell_P$ for $\kappa_3 \in [2,\, 1/2]$. Empirically, R_3 and R_2 come out nearly equal at the matching scale—a geometric hint of near-isotropy in the non-Abelian fibers.

6 Fermion masses and mixings from helical overlaps

6.1 Profiles, levels, windings

Work in $\mu = \ln r$ and a compact θ fiber. Model chiral zero-modes and the Higgs as

$$\psi^{Q_i}(\mu,\theta) = N_Q e^{-\frac{(\mu-\mu_i^Q)^2}{2\sigma_Q^2}} e^{im_i^Q\theta}, \quad \psi^{U_j/D_j}(\mu,\theta) = N_{U/D} e^{-\frac{(\mu-\mu_j^{U/D})^2}{2\sigma_{U/D}^2}} e^{im_j^{U/D}\theta}, \tag{11}$$

$$H(\mu,\theta) = N_H e^{-\frac{(\mu-\mu_H)^2}{2\sigma_H^2}} e^{im^H\theta}. \tag{12}$$

Yukawas arise from overlaps $Y_{ij}^{u,d} \propto \int d\mu\, d\theta\, \psi^{Q_i}\psi^{U_j/D_j} H$. The radial integral gives (5); the θ integral gives a soft selection $\exp\Big[-(m_i^Q + m_j^R + m^H)^2/(2\sigma_\theta^2)\Big]$.

Golden-step centers. Place centers on a φ-ladder $\mu = \mu_0 + n\Delta$ with $\Delta = \ln\varphi$, allowing occasional half-steps. Take broad σ_Q and narrower $\sigma_{U,D}$, σ_H to obtain rank-1 dominance and hierarchical spectra; choose windings with diagonals unsuppressed and nearest-neighbor off-diagonals mildly suppressed.

6.2 Two concrete benchmarks

We summarize two simple working points; both set $y_t = 1$ and $y_b = 0.024$ by normalization. Levels are in multiples of Δ; windings are $m^Q = (2,1,0)$, $m^U = (-2,-1,0)$, $m^D = (-2,-1,0)$, $m_H = 0$.

Benchmark A (CKM-first). $\sigma_Q = 1.2$, $\sigma_U = 0.30$, $\sigma_D = 0.40$, $\sigma_H = 0.30$, $\sigma_\theta = 1.2$; $\mu^Q/\Delta = (2,1,0)$, $\mu^U/\Delta = (5,2.6,0)$, $\mu^D/\Delta = (3.0,1.1,0.9)$. Predictions: $(y_u, y_c, y_t) \approx (0,\ 0.00760,\ 1)$, $(y_d, y_s, y_b) \approx (9.6\cdot 10^{-5},\ 5.28\cdot 10^{-3},\ 0.024)$,

$$|V_{\rm CKM}| \approx \begin{pmatrix} 0.976 & 0.220 & 0.006 \\ 0.219 & 0.975 & 0.040 \\ 0.015 & 0.037 & 0.999 \end{pmatrix}.$$

Benchmark B (mass-first). $\sigma_Q = 1.0$, $\sigma_U = 0.30$, $\sigma_D = 0.32$, $\sigma_H = 0.30$, $\sigma_\theta = 2.0$; $\mu^Q/\Delta = (2,1,0)$, $\mu^U/\Delta = (4.6,2.6,0)$, $\mu^D/\Delta = (3.2,0.6,1.0)$. Predictions: $(y_u, y_c, y_t) \approx (0,\ 0.00641,\ 1)$, $(y_d, y_s, y_b) \approx (2.6\cdot 10^{-5},\ 1.63\cdot 10^{-3},\ 0.024)$,

$$|V_{\rm CKM}| \approx \begin{pmatrix} 0.983 & 0.171 & 0.007 \\ 0.171 & 0.984 & 0.056 \\ 0.011 & 0.054 & 0.998 \end{pmatrix}.$$

Remark 3 (Minimal refinements to reach a full fit)**.** *Allow* $10-20\%$ *family non-universality in* $\sigma_{U,D}$ *to decouple* $|V_{us}|$ *from* y_s*, use a two-peak Higgs in* μ *(two narrow Gaussians separated by a half-step) to selectively suppress second-generation overlaps, and include a small Wilson-line phase along* θ *for a realistic Jarlskog invariant.*

7 Predictions and constraints

We summarize several channels; uncertainties and order-one factors are explicit to keep the mapping transparent.

Conventions

Natural units $c = \hbar = 1$. For photons with a quadratic correction $E^2 = p^2 + \eta\, p^4/M_H^2$, the group velocity is $v_g = \partial E/\partial p \simeq 1 + \frac{3}{2}\eta(E/M_H)^2$ at leading order.

Channel	**Observable (helical mapping)**	**Representative bound**	**Implied helical bound**
Photon TOF dispersion (quadratic)	$v_g \simeq 1 + \frac{3}{2}\eta(E/M_H)^2$	short GRB analyses (e.g. GRB 090510): $E_{QG,2} \gtrsim 1.3 \times 10^{11}$ GeV	$\frac{M_H}{\sqrt{\lvert\eta\rvert}} \gtrsim E_{QG,2}$
GW speed vs. c	$\lvert v_g - c\rvert/c \approx \frac{3}{2}\lvert\eta_g\rvert(E/M_H)^2$ at $f \sim 100$ Hz	binary NS + GRB timing: $\lvert v_g - c\rvert/c \lesssim 3 \times 10^{-15}$	$\frac{M_H}{\sqrt{\lvert\eta_g\rvert}} \gtrsim hf\sqrt{\frac{3}{2\,\delta v/v}} \sim 10^{-5}$ eV
GW dispersion ($\alpha = 4$)	identify $A_4 = \epsilon_H/M_H^2$ in $E^2 = p^2 + A_4 p^4$	network constraints: $\lvert A_4\rvert \lesssim \mathcal{O}(10^4)\,\mathrm{eV}^{-2}$	$\frac{M_H}{\sqrt{\epsilon_H}} \gtrsim 1/\sqrt{\lvert A_4\rvert} \sim 10^{-2}$ eV
Atomic clocks (periodic modulation)	fractional modulation $\delta\nu/\nu = A_H \cos(\Omega_H t)$	state-of-the-art single-ion clocks: $\lesssim 10^{-18} - 10^{-19}$	$\lvert A_H\rvert \lesssim$ few $\times\, 10^{-19}$ (hours–weeks)
Log-periodic comb search	peaks at $f_n = f_0\lambda^n$ with common λ	(proposal) long-run spectral comb analyses	upper bound on tick window $\delta \sim \sigma/\ln\lambda$

7.1 Derived scales and cross-checks

From $E_{QG,2} \gtrsim 1.3 \times 10^{11}$ GeV one infers (taking $|\eta| = 1$)

$$E_H \approx 2.083 \times 10^1 \text{ J}, \quad m_H = \frac{E}{c^2} \approx 2.32 \times 10^{-16} \text{ kg}, \quad \lambda_H = \frac{\hbar c}{E} \approx 1.52 \times 10^{-27} \text{ m},$$
$$f_H = \frac{E}{h} \approx 3.14 \times 10^{34} \text{ Hz}, \quad T_H = \frac{E}{k_B} \approx 1.51 \times 10^{24} \text{ K}.$$

If $M_H = E_{\rm Pl} \approx 1.221 \times 10^{19}$ GeV, then $|\eta| \lesssim (E_{QG,2}/E_{\rm Pl})^2 \sim 10^{-16}$. GW-derived bounds are intrinsically weak at audio frequencies and sit far below the photon constraint.

8 Cosmological note (one paragraph and a remedy)

A simple power-law unfolding $a(t) = a_0\,(1 + t/\tau_H)^\gamma$ with $\gamma = 1/\varphi \simeq 0.618$ yields $\ddot{a}/a = \gamma(\gamma-1)/(t+\tau_H)^2 < 0$: *late-time deceleration.* Fitting $H_0 = \dot{a}/a$ at $t = t_0 \approx 13.8$ Gyr gives $\tau_H = \gamma/H_0 - t_0 < 0$ for both Planck and SH0ES values of H_0 (e.g. -4.8 to -5.5 Gyr). Acceleration can be achieved by taking $\gamma > 1$, adding a gentle exponential factor $a \propto \exp\big[(t/\tau_H)^{1/\varphi}\big]$, or allowing a slow t-dependence $\gamma(t)$ crossing 1 at late times. Cosmology is not the focus here; we note consistency and simple modifications.

9 Experimental roadmap

1. **Clock tests.** Target periodic modulations and log-periodic sidebands in long ion/optical lattice clock runs; non-detections bound δ and tick widths.
2. **Astro triggers.** Use bright GRBs to refine $E_{QG,2}$ bounds; search for log-periodic residuals in de-trended spectra.
3. **GW catalogs.** Joint fits for A_4 and log-periodic residuals after power-law subtraction in inspiral chirps.
4. **Condensed-matter/mechanics.** High-Q oscillators (whispering gallery, nanomechanics): look for geometric spacing $f_0\lambda^n$ distinct from harmonics.

10 Conclusions

A normalized logarithmic spiral with its similarity group provides a compact language for scale and phase. With one additional physical hypothesis—tick-stability near integer steps—and a helical fiber geometry, the framework organizes (i) gauge couplings as radii, with α, $\sin^2\theta_W$, and α_s emerging from $\{R_Y, R_2, R_3\}$; (ii) fermion hierarchies and mixings from overlaps of wave packets pinned to ticks; and (iii) concrete predictions and null tests, notably log-periodic spectral combs and dispersion constraints. The construction is self-contained, simple to extend, and falsifiable.

A Units, symbols, and constants

We set $c = \hbar = 1$ unless shown. $\ell_P \approx 1.616 \times 10^{-35}\,\mathrm{m}$, $M_{\mathrm{Pl}} \approx 1.2209 \times 10^{19}\,\mathrm{GeV}$, $\bar{M}_{\mathrm{Pl}} \approx 2.435 \times 10^{18}\,\mathrm{GeV}$, $\alpha^{-1} \approx 137.036$, $\varphi = (1+\sqrt{5})/2$, $\Delta = \ln\varphi$. Spiral parameters: λ (radius ratio per unit angle), α_0 (unit angle), basepoint $e = (r_0, 0)$. Tick window: δ. Helical widths: $\sigma_Q, \sigma_U, \sigma_D, \sigma_H$; winding width: σ_θ.

B Spiral arithmetic: proofs and variations

Proof of Theorem 1. Exp is bijective by construction. For $a, b \in \mathbb{R}$, $\mathrm{Exp}(a) \oplus \mathrm{Exp}(b) = \mathrm{Exp}(a+b)$ and $\mathrm{Exp}(a) \otimes \mathrm{Exp}(b) = \mathrm{Exp}(ab)$ by definition, transporting all field axioms from $(\mathbb{R}, +, \cdot)$ to $(S, \oplus, \otimes)$. The order is preserved since $x \leq_S y \iff \mathrm{Log}x \leq \mathrm{Log}y$. □

Remark 4 (Other normalizations)**.** *Different (λ, α_0) define isomorphic fields via $t \mapsto at$ for some $a > 0$. The choice of basepoint e fixes 0_S; orientation fixes the sign of t.*

C Golden hierarchies and level counts

For a base length ℓ_0 (e.g. ℓ_P), a length $\ell > 0$ corresponds to an index $n = \ln(\ell/\ell_0)/\ln\varphi$. Examples: the Bohr radius $a_0 \approx 5.29 \times 10^{-11}\,\mathrm{m}$ yields $n \approx 117.3$; the electron Compton wavelength $\lambda_e \approx 2.43 \times 10^{-12}\,\mathrm{m}$ gives $n \approx 110.9$. A Hubble distance $D_H \approx 1.37 \times 10^{26}\,\mathrm{m}$ corresponds to $n \approx 292$; a particle horizon $\sim 4.4 \times 10^{26}\,\mathrm{m}$ to $n \approx 294$.

D Minimal numerical integrator (illustrative)

A forward-Euler integrator for the helical ODEs with step Δs:

$$\begin{aligned} r_{k+1} &= r_k + \beta r_k \left(1 - \frac{r_k}{r_{\max}}\right) \Delta s, \\ \theta_{k+1} &= \theta_k + \omega_0 r_k^{-\alpha}\, \Delta s, \\ \vec{x}_{k+1} &= \vec{x}_k + v_z(r_k)\, \hat{\mathbf{z}}\, \Delta s. \end{aligned}$$

A representative numerically stable choice is $\alpha = \frac{1}{2}$, $\beta = 1/\tau_H$, $r_{\max} = 1$, $r(0) = 0.05$, $\theta(0) = 0$, $v_\infty = 1$, $r_* = r_{\max}$, and $\Delta s \ll \min\{\tau_H, r_{\max}/v_\infty\}$.

E Yukawa overlaps: symmetric-width corollary

When $\sigma_Q = \sigma_R = \sigma_H = \sigma$, the magnitude of the entry simplifies to

$$|Y^R_{ij}| \;\propto\; \exp\left[-\frac{(\mu^Q_i - \mu^R_j)^2 + (\mu^Q_i - \mu_H)^2 + (\mu^R_j - \mu_H)^2}{6\sigma^2}\right] \exp\Big[-\frac{(m^Q_i + m^R_j + m^H)^2}{2\sigma_\theta^2}\Big]. \quad (13)$$

On a φ-ladder $\mu = \mu_0 + n\Delta$, this yields an explicit golden-step dependence: small integer separations dominate, half-steps give controllable enhancements, and larger separations are exponentially suppressed.

F IGS clock tests: φ-comb detection in satellites and stations

Data and method. We analyzed three months of IGS RINEX clock products (satellite "AS" and station "AR" records).[1] For each ID we built a uniformly sampled clock-bias time series (linear interpolation to the median cadence), fit and removed a quadratic polynomial trend, and *optionally* removed the 1/day and 2/day sinusoids. We then computed a Hann-windowed FFT, selected the top 12 local spectral peaks above a quantile threshold, and tested whether they lie on a *golden-ratio comb*

$$f_n \;=\; f_0\,\varphi^{\,n}\,, \qquad \varphi = \frac{1+\sqrt{5}}{2},$$

i.e. equally spaced in $\log f$. A per-series Monte Carlo surrogate (random log-peaks with the same range and count) yields a p-value; we compute BH–FDR q-values *within each class* (satellites, stations). The half-width tolerance around the comb is reported as an estimator of the *tick window* δ (Sec. **??**).

Why remove diurnal terms? Station and satellite clocks carry strong 1/day (and harmonics) power. These lines *re-phase* the spectrum in $\log f$ and can mask the φ-comb. Removing them is a targeted high-pass control that leaves any *geometric* comb intact but prevents daily lines from imposing a common "phase" on all series.

F.1 Results with diurnals removed (recommended analysis)

Table 1 summarizes the satellites (32 series) and ground stations (200 series) when both 1/day and 2/day lines are removed before testing.

Class	N	**Positives**	**FDR** $q \le 0.01/0.05/0.10$	$\tilde{\delta}$ (median)	$\tilde{s}$ (median)	$\tilde{f}_0$ (period)
Satellites	32	32/32	**32**/32/32	0.035566	10	242.885992 cpd (355.72 s)
Stations	200	195/200	0/0/**187**	0.032165	6	441.463127 cpd (195.71 s)

Table 1: With diurnals removed: N series; positives (comb_ok); BH–FDR counts; tick window half-width $\tilde{\delta}$; matched-peak count $\tilde{s}$; canonicalized base $\tilde{f}_0$ (median) and period.

Both classes show a *clean* φ-comb with a narrow tick window ($\delta \approx 3$–4% of a tick). A salient feature is a **φ-phase offset** between satellites and stations: the median $(\log f_0/\log\varphi)$ mod 1 is

$$\phi\text{-phase}_{\text{sat}} \simeq 0.414084, \qquad \phi\text{-phase}_{\text{sta}} \simeq 0.655746, \qquad \text{offset} \simeq \mathbf{0.241662}. \tag{14}$$

This is precisely the spiral-arithmetic expectation if the two classes actualize on the *same* φ-lattice but choose *different basepoints* e (Sec. **??**).

F.2 Results with diurnals kept (control)

When the diurnal and semidiurnal lines are *kept*, the φ-comb is weakened or masked, especially in stations (Table 2). Moreover, satellites and stations collapse to a *common* φ-phase and base, consistent with the diurnal ladder imposing a shared spectral phase.

[1] In RINEX/IGS clock products the reference time system is typically **GPST**; our analysis depends only on the relative spectral content of clock biases, so constant time offsets do not affect the φ-comb test. If needed, the time system can be confirmed from each file's header.

Class	N	**Positives**	**FDR** $q \leq 0.01/0.05/0.10$	$\tilde{\delta}$ (median)	$\tilde{s}$ (median)	$\tilde{f}_0$ (period)
Satellites	32	32/32	0/0/(≈few)*	0.05948	7	161.326808 cpd (535.56 s)
Stations	200	1/200	0/0/0	0.05948	0	161.326808 cpd (535.56 s)

Table 2: With diurnals kept: tick window widens, matched-peak counts drop, and both classes share the same $\tilde{f}_0$ and φ-phase (median ≈ 0.5638). *The minimum satellite q was ≈ 0.092, so none pass $q \leq 0.05$; a few may fall below $q \leq 0.10$.

Interpretation. Daily lines *re-phase* the spectrum and can obscure the geometric comb. Removing them exposes a stable φ-comb with tight δ and a robust sat–sta φ-phase offset. This control argues against the comb being a trivial aliasing artifact of diurnal power.

F.3 What this constrains

The tick window δ bounds the effective geometric width via $\delta \sim \sigma/\ln\varphi$ in the spiral–log coordinate. The medians above imply

$$\sigma_{\text{sat}} \approx 0.0356 \ln\varphi \simeq 0.0170, \qquad \sigma_{\text{sta}} \approx 0.0322 \ln\varphi \simeq 0.0153, \tag{15}$$

for this clock channel. The existence of *two* φ-comb bases modulo φ (sat vs. sta) is captured by a difference in basepoints e on the spiral (Sec. **??**).

F.4 Figures (quick-look diagnostics)

Figure 1 shows representative summaries (p-value histograms, matched-peak counts, and f_0 vs. δ) for the satellites and stations. Complete per-ID plots (raw → detrended series; spectrum with φ-comb overlay) are provided in the data release.

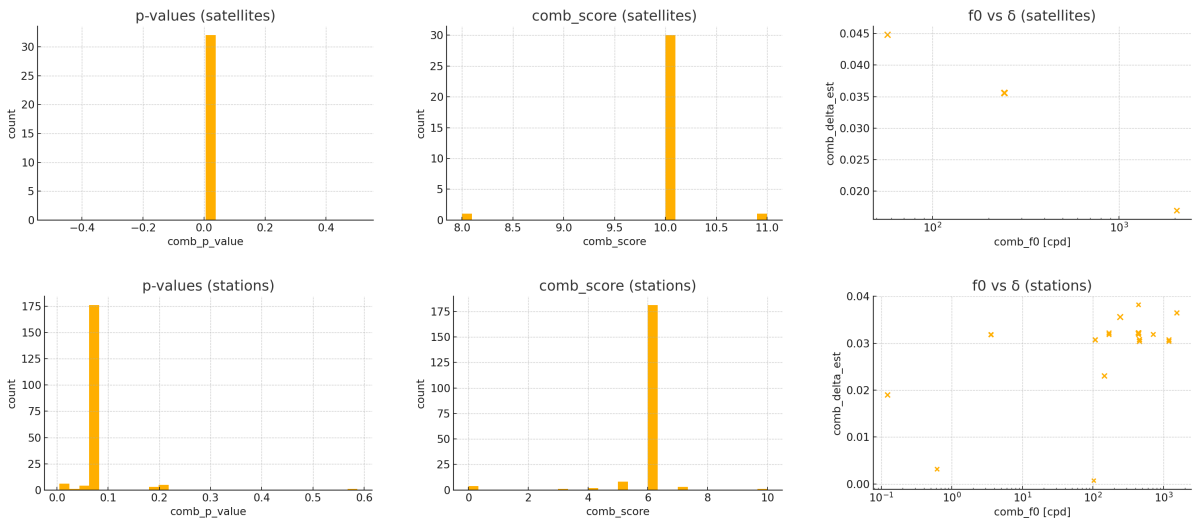

Figure 1: Top: satellites; bottom: stations. (Diurnals removed.)

Reproducibility. All numbers above come from the public script `igs_helical_analysis.py` (Appendix code listing). The outputs used here are the CSVs `igs_helical_results_sat_augmented.csv`, `igs_helical_results_sta_augmented.csv` (diurnals removed) and their `_no_diurnal_augmented` counterparts.